THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 90	At full hours the minute hand is at 12; each hour mark is 30°.
2	**(a)** acute	Classify the turn.
3	**(b)** 75	Add the adjacent angles.
4	**(d)** 270	Remaining turn $= 360^\circ -$ turn.
5	**(a)** 6	Each pair of rays makes an angle.
6	**(d)** 145	Inner and outer scales add up to 180°.
7	**(d)** collinear points	Definition / property
8	**(b)** 180	At full hours the minute hand is at 12; each hour mark is 30°.
9	**(c)** right	Classify the turn.
10	**(a)** 80	Add the adjacent angles.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Adjacent angles add up (1); $46 + 38 = 84^\circ$ (1).	1+1
12	Hour hand points to 3 on the dial; each mark is 30° (1); Angle $= 90^\circ$ (1).	1+1
13	Pairs of points (1); 6 segments (1).	1+1
14	Adjacent angles add up (1); $59 + 41 = 100^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Every 90° is a quarter turn: N to E to S to W (1); 90° gives East (1); Type: right (1).	1+1+1
16	Draw a ray OA, place the centre of the protractor at O with the zero line along OA (1); Mark a point at 40° and join O to it (1); The angle is acute (1).	1+1+1
17	One right angle is 90° (1); $180 \div 90 = 2$ (1); 3 right angles make 270° (1).	1+1+1
	OR: Every 90° is a quarter turn: N to E to S to W (1); 180° gives South (1); Type: straight (1).	1+1+1
18	Draw a ray OA, place the centre of the protractor at O with the zero line along OA (1); Mark a point at 75° and join O to it (1); The angle is acute (1).	1+1+1
	OR: One right angle is 90° (1); $270 \div 90 = 3$ (1); 5 right angles make 450° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\angle AOC = 110 - 30 = 80^\circ$ (1); $\angle AOB = 30^\circ$: acute (1); $\angle BOC = 110^\circ$: obtuse (1); $\angle AOC = 80^\circ$: acute (1); State the arms and vertex (1).	1+1+1+1+1
	OR: The hour hand turns 30° every hour (1); Hours passed $= 4 - 1 = 3$, hour hand turn $= 90^\circ$ (1); Minutes passed $= 180$; the minute hand turns 3 full turns $= 1080^\circ$ (1); Type of the hour hand's turn: right (2).	1+1+1+2
20	$\angle AOC = 135 - 45 = 90^\circ$ (1); $\angle AOB = 45^\circ$: acute (1); $\angle BOC = 135^\circ$: obtuse (1); $\angle AOC = 90^\circ$: right (1); State the arms and vertex (1).	1+1+1+1+1
	OR: The hour hand turns 30° every hour (1); Hours passed $= 5 - 2 = 3$, hour hand turn $= 90^\circ$ (1); Minutes passed $= 180$; the minute hand turns 3 full turns $= 1080^\circ$ (1); Type of the hour hand's turn: right (2).	1+1+1+2

Mistakes to avoid:

- Reading the wrong scale on the protractor (inner vs outer). Decide first whether the angle is acute or obtuse.
- Using 30° per minute for the minute hand; it is 6° per minute, and 30° per hour is for the hour hand.
- Calling a 200° or 225° angle obtuse. Obtuse angles are less than 180°; larger ones are reflex.
- Saying a ray has two end points; a ray has exactly one end point.
- Counting line segments for n points as $n(n-1)$ without dividing by 2 (segment AB and BA are the same).

- Placing the protractor's centre off the vertex, or starting the scale from the wrong arm.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 225	Remaining turn $= 360^\circ -$ turn.
2	**(c)** 10	Each pair of rays makes an angle.
3	**(b)** 125	Inner and outer scales add up to 180°.
4	**(d)** initial point (vertex)	Definition / property
5	**(d)** 90	At full hours the minute hand is at 12; each hour mark is 30°.
6	**(b)** obtuse	Classify the turn.
7	**(b)** 100	Add the adjacent angles.
8	**(a)** 135	Remaining turn $= 360^\circ -$ turn.
9	**(c)** 15	Each pair of rays makes an angle.
10	**(c)** 110	Inner and outer scales add up to 180°.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Hour hand points to 5 on the dial; each mark is 30° (1); Angle $= 150^\circ$ (1).	1+1
12	Pairs of points (1); 10 segments (1).	1+1
13	Adjacent angles add up (1); $73 + 62 = 135^\circ$ (1).	1+1
14	Hour hand points to 10 on the dial; each mark is 30° (1); Angle $= 60^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Every 90° is a quarter turn: N to E to S to W (1); 270° gives West (1); Type: reflex (1).	1+1+1
16	Draw a ray OA, place the centre of the protractor at O with the zero line along OA (1); Mark a point at 100° and join O to it (1); The angle is obtuse (1).	1+1+1
17	One right angle is 90° (1); $360 \div 90 = 4$ (1); 6 right angles make 540° (1).	1+1+1
	OR: Every 90° is a quarter turn: N to E to S to W (1); 135° gives between East and South (1); Type: obtuse (1).	1+1+1
18	Draw a ray OA, place the centre of the protractor at O with the zero line along OA (1); Mark a point at 125° and join O to it (1); The angle is obtuse (1).	1+1+1
	OR: One right angle is 90° (1); $450 \div 90 = 5$ (1); 2 right angles make 180° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\angle AOC = 90 - 25 = 65^\circ$ (1); $\angle AOB = 25^\circ$: acute (1); $\angle BOC = 90^\circ$: right (1); $\angle AOC = 65^\circ$: acute (1); State the arms and vertex (1).	1+1+1+1+1
	OR: The hour hand turns 30° every hour (1); Hours passed $= 9 - 3 = 6$, hour hand turn $= 180^\circ$ (1); Minutes passed $= 360$; the minute hand turns 6 full turns $= 2160^\circ$ (1); Type of the hour hand's turn: straight (2).	1+1+1+2
20	$\angle AOC = 170 - 60 = 110^\circ$ (1); $\angle AOB = 60^\circ$: acute (1); $\angle BOC = 170^\circ$: obtuse (1); $\angle AOC = 110^\circ$: obtuse (1); State the arms and vertex (1).	1+1+1+1+1
	OR: The hour hand turns 30° every hour (1); Hours passed $= 8 - 5 = 3$, hour hand turn $= 90^\circ$ (1); Minutes passed $= 180$; the minute hand turns 3 full turns $= 1080^\circ$ (1); Type of the hour hand's turn: right (2).	1+1+1+2

Mistakes to avoid:

- Reading the wrong scale on the protractor (inner vs outer). Decide first whether the angle is acute or obtuse.
- Using 30° per minute for the minute hand; it is 6° per minute, and 30° per hour is for the hour hand.
- Calling a 200° or 225° angle obtuse. Obtuse angles are less than 180°; larger ones are reflex.
- Saying a ray has two end points; a ray has exactly one end point.
- Counting line segments for n points as $n(n-1)$ without dividing by 2 (segment AB and BA are the same).

- Placing the protractor's centre off the vertex, or starting the scale from the wrong arm.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** arms	Definition / property
2	**(b)** 60	At full hours the minute hand is at 12; each hour mark is 30°.
3	**(a)** straight	Classify the turn.
4	**(b)** 115	Add the adjacent angles.
5	**(b)** 90	Remaining turn $= 360^\circ -$ turn.
6	**(b)** 21	Each pair of rays makes an angle.
7	**(a)** 70	Inner and outer scales add up to 180°.
8	**(c)** 120	At full hours the minute hand is at 12; each hour mark is 30°.
9	**(d)** reflex	Classify the turn.
10	**(b)** reflex	Classify the turn.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Pairs of points (1); 15 segments (1).	1+1
12	Adjacent angles add up (1); $28 + 87 = 115^\circ$ (1).	1+1
13	Hour hand points to 8 on the dial; each mark is 30° (1); Angle $= 120^\circ$ (1).	1+1
14	Pairs of points (1); 28 segments (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Every 90° is a quarter turn: N to E to S to W (1); 225° gives between South and West (1); Type: reflex (1).	1+1+1
16	Draw a ray OA, place the centre of the protractor at O with the zero line along OA (1); Mark a point at 160° and join O to it (1); The angle is obtuse (1).	1+1+1
17	One right angle is 90° (1); $540 \div 90 = 6$ (1); 7 right angles make 630° (1).	1+1+1
	OR: Every 90° is a quarter turn: N to E to S to W (1); 315° gives between West and North (1); Type: reflex (1).	1+1+1
18	Draw a ray OA, place the centre of the protractor at O with the zero line along OA (1); Mark a point at 210° and join O to it (1); The angle is reflex (1).	1+1+1
	OR: One right angle is 90° (1); $630 \div 90 = 7$ (1); 8 right angles make 720° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\angle AOC = 150 - 40 = 110^\circ$ (1); $\angle AOB = 40^\circ$: acute (1); $\angle BOC = 150^\circ$: obtuse (1); $\angle AOC = 110^\circ$: obtuse (1); State the arms and vertex (1).	1+1+1+1+1
	OR: The hour hand turns 30° every hour (1); Hours passed $= 10 - 4 = 6$, hour hand turn $= 180^\circ$ (1); Minutes passed $= 360$; the minute hand turns 6 full turns $= 2160^\circ$ (1); Type of the hour hand's turn: straight (2).	1+1+1+2
20	$\angle AOC = 120 - 55 = 65^\circ$ (1); $\angle AOB = 55^\circ$: acute (1); $\angle BOC = 120^\circ$: obtuse (1); $\angle AOC = 65^\circ$: acute (1); State the arms and vertex (1).	1+1+1+1+1
	OR: The hour hand turns 30° every hour (1); Hours passed $= 11 - 6 = 5$, hour hand turn $= 150^\circ$ (1); Minutes passed $= 300$; the minute hand turns 5 full turns $= 1800^\circ$ (1); Type of the hour hand's turn: obtuse (2).	1+1+1+2

Mistakes to avoid:

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- Saying a ray has two end points; a ray has exactly one end point.
- Counting line segments for n points as $n(n-1)$ without dividing by 2 (segment AB and BA are the same).

- Placing the protractor's centre off the vertex, or starting the scale from the wrong arm.

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